Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Orbit above the equator **Or** above the same point [on the ground]	1			1		
	(b)	(i)		Substitution: $\frac{36\,000\,000}{3 \times 10^8}$ (1) = 0.12 [s] (1) Doubling to 0.24 [s] (1) Award 2 marks for answer of 2.4×10^n where $n \neq -1$ Award 1 mark for answer of 1.2×10^n where $n \neq -1$ **Alternative:** $\frac{72\,000\,000\ (1)}{3 \times 10^8}$ (1) = 0.24 [s] (1)	1	1 1		3	3	
		(ii)		Substitution: $3 \times 10^8 = f \times 0.002$ (1) Manipulation: $f = \frac{3 \times 10^8}{0.002}$ (1) $= 1.5 \times 10^{11}$ [Hz] (1) Award 2 marks if the wrong wavelength is used i.e. 1 m then $f = 3 \times 10^8$ [Hz]	1	1 1		3	3	
		(iii)		Radio [waves]	1			1		
				Question 5 total	**4**	**4**	**0**	**8**	**6**	**0**